

Yang Wu

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Research Interests

Imagination and Mental Simulation, with a focus on the role of memory recombination and memory retrieval.

Education

Soochow University	2023 – 2027
<i>B.S. in Mathematics & B.A. in Music Composition</i>	<i>Taipei City, Taiwan</i>
Neuropsychiatric Disorder Summer Associate School (IBRO)	2024
<i>National Taiwan University</i>	<i>Taipei City, Taiwan</i>
Cognitive Neuroscience Summer School	2024
<i>National Central University</i>	<i>Taoyuan City, Taiwan</i>
Florida State University	2022 – 2023
<i>B.A. in Music (Piano Performance)</i>	<i>Tallahassee, FL, USA</i>

Research Experience

Marcelo Mattar lab, New York University	2025/7 – Present
<i>Department of Psychology</i>	<i>New York, NY</i>
<i>Advisor: Marcelo Mattar & Daniel Levenstein (Yale)</i>	
• Planning via Cognitive Map Learned in Predictive RNN (with Daniel Levenstein): Coupling the rollout-based planning network (Jensen, Hennequin & Mattar, 2024) to the predictive RNN whose place-cell-like representations, theta sweeps, and replay emerge from sequential predictive learning (Levenstein et al., 2024), to test whether planning via cognitive map learned from partially observable sequential predictive task improve navigation behavior. • Designed the combined architecture (frozen predictive world model + recurrent controller that imagines k-step action sequences and integrates the rolled-out trajectory) and the control experiments to verify the functionality of theta sweep rollouts in the model. • Modeling Episodic Simulation through Memory Recombination: Built key-value episodic memory module with a GRU controller and a three-phase reinforcement learning task (exploration → mental simulation → execution) to model how humans recombine past episodes for future planning (Schacter et al., 2007); established that Phase 2 mental simulation improves Phase 3 execution.	
James Whittington lab, University of Oxford	2025/7 – Present
<i>Department of Experimental Psychology</i>	<i>Oxford, UK</i>
<i>Advisors: James Whittington & Joseph Warren</i>	
• Cognitive Map Reorganization in Sequential Memory Task: Identified specific algorithms the RNN uses in sequential memory task, extending previous work “A tale of two algorithms” (Whittington et al., 2024) to model how the hippocampus reorganizes its cognitive map during sequence learning. • Neural Subspace Stretching as a Representational Signature of Sequence Learning in RNN: Used linear decoders to test competing hypotheses about observation encoding in RNN. Identified the “stretching” of neural subspaces during sequence learning of RNN that generalized across multiple loop lengths, and quantified different learning phases’ decoder alignment via cosine similarity.	
Kanaka Rajan lab, Harvard University	2026/2 – 2026/7
<i>Harvard Medical School / Kempner Institute</i>	<i>Boston, MA</i>
<i>Advisor: John Vastola</i>	
• Characteristics of Multi-Timescale Efficient-Coding Neural Network under Energy Efficiency Constraint: Designed and ran Contrastive Predictive Coding experiments with InfoNCE as the Mutual Information in the Efficient-Coding hypothesis, confirming three theoretical predictions about how representation timescales emerge in Efficient-Coding networks with temporal smoothness energy costs.	

Kanaka Rajan lab, Harvard University

Harvard Medical School / Kempner Institute

Advisors: Ryan Badman & John Vastola

2025/7 – 2025/12

Boston, MA

- **Interpreting Planning-Like Behavior in Naturalistic Foraging RL Agents:** Analyzed emergent planning-like behaviors ethologically in model-free reinforcement learning agents within naturalistic environments.
- Implemented Sparse Autoencoder, Generalized Linear Model, Decoder and Auto-Correlation to probe agents' internal representations and external behavior.
- Contributed code and analysis to *Deep RL Needs Deep Behavior Analysis: Exploring Implicit Planning by Model-Free Agents in Open-Ended Environments, NeurIPS 2025*; formally recognized in acknowledgements.

Yu-Wei Wu lab, Academia Sinica

Institute of Molecular Biology

Advisors: Yu-Wei Wu & Yiko Chen

2024/7 – 2025/7

Taipei, Taiwan

- **Decomposing Inter-Regional Neural Dynamics during Mouse Motor Control:** Used Current-Based Decomposition Method (CURBD) and Dissociative Prioritized Analysis of Dynamics (DPAD) to model and describe inter-regional communication between M1, M2 and S1 during mouse on-cue reaching task.

Yu-Te Wang lab, Academia Sinica

Research Center for Information Technology Innovation

Advisor: Yu-Te Wang

2024/2 – 2025/3

Taipei, Taiwan

- **Development of 3D Printed Semi-Dry EEG Electrodes:** Built EEG electrodes with 3D printing; collected human EMG/EEG data; applied CompactCNN for EEG analysis. Led to multiple conference and IEEE journal publications.
- VR/AR-Based HCI Telerehabilitation Systems for EMG Motor Recovery using Unity.

Publications

3D printed watermill-like semi-dry electrodes for BCI applications. IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2026.

C.-M. Chung, Y.-T. Wang*, J.-B. Lu, Y.-C. Hsu, Y.-J. Su, **Y. Wu**, C.-F. Hung

Posters

HCI Technologies in rehabilitation: enhancing physical therapy with sEMG, VR and LLM-Generated music. Society for Neuroscience Annual Meeting, 2024.

Y. Wu*, Y.-J. Su, J.-B. Lu, Y.-T. Wang*, T.-H. Cheong

RehabVerse: Mixed Reality Solution for Telerehabilitation. National Biotechnology Research Park Pitch Day, 2024.

Y. Wu*, Y.-J. Su, J.-B. Lu, Y.-H. Chen, Y.-T. Wang*

BrainPrint: Innovative Head-Mounted EEG Technology for Secure Personal Identification. IEEE Brain Discovery & Neurotechnology Workshop, 2024.

Y.-L. Chu, C.-C. Tsao, C.-M. Chung, C.-F. Hung, Y.-Y. Lee, J.-B. Lu, **Y. Wu**, Y.-H. Chen, Y.-T. Wang*

Awards, Grants & Extracurricular

Ministry of Education Study Abroad Programme for Future Scholars in Humanities and Social Sciences (US\$50,000)	2026
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Computational and Systems Neuroscience Conference (COSYNE) Undergraduate Travel Grant (US\$1,500)	2025
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Chu-Chin Hsu Scholar	2025
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Dean's List (Academic Excellence Award) × 2	2025
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Distinction Award, International ICT Innovation Services Award (BrainPrint)	2024
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National Science and Technology Council Neuroscience Camp	2024
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Dean's List (Academic Excellence Award)	2024
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Academia Sinica Statistical Science Camp	2024
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Finalist, National Biomedical Engineering Award (BrainPrint)	2024
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Courses & Skills

Selected Courses (grades on 100-point scale): **Computational Neuroscience (Grad Level) 95, Statistical Mechanics (Grad Level) 87**, Advanced Calculus (Real Analysis) 99, Linear Algebra 96, Programming in Life Science 97, Life Science (Physiology) 95.

Skills: Python; Bayesian inference; Deep learning (PyTorch, RNNs, Transformers); Reinforcement learning (model-free and model-based, PPO, GRPO); Statistical modeling (GLMs, sparse autoencoders, linear decoders); Data collection (EEG, EMG); Tools (Git, Blender, MATLAB, 3D printing).

Music

Milonga – Latin-American piece for Cello and Piano. vimeo.com/1036267350

Zhang Beihai said good-bye to his father – Solo Piano. vimeo.com/1036267291

Weathering With You Movie Trailer (Re-score). vimeo.com/1036267308